

Statistical Analysis of Root-Hormone: Effect of Phytohormones on Plant Root Lengths and the Correlations of Change

Rita Lahaj (I6382311)

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Introduction

Hormones in plants, also referred to as phytohormones, largely influence growth of the organism. [1] They span different functional classes, such regulators of elongation and dormancy mechanisms. The most central phytohormone, *auxin* (IAA), largely controls a plant's growth in response to environmental factors. [2] *Cytokinin* (CK) also plays a crucial role in regulation of cell division and metabolism. [3] Despite both of the aforementioned hormones reacting to different abiotic and biotic stressors themselves, *abscisic acid* (ABA) is essential for a plant's response to external stress, controlling downstream pathways following environmental changes. [4]

Alongside other production sites, all three hormones are produced in roots. The root, denoting the typically underground part of the plant, works to anchor the organism and acquire nutrients from the soil. It also acts as a reserve, storing excess carbohydrates produced by photosynthesis or extra water in preparation for environmental stress such as drought. [5] As such, the roots are often referred to as the "heart" of the plant. Despite this, there is rarely research done with a focus on this structure.

This report poses the research question: "How strongly does each hormone change total root length compared to control plants, and do the three hormones differ from each other?". Datasets from AraPheno are used in an attempt to answer this question. [6, 7, 8, 9, 10] Statistical analysis is completed through C++, and visualizations of results through Python.

Data sample

The datasets utilized herein were sourced from AraPheno. AraPheno refers to the phenotypes, or expressed traits, of *Arabidopsis thaliana*, a well-studied small plant that is often used as a model organism in genetic studies. [11] The datasets include the mean root length of samples grown in control (untreated by any hormone), treated with IAA, treated with CK, and treated with ABA. The code of this research uses 3 columns from the datasets: **accession_id**, **phenotype_value**, and **phenotype_name**. **accession_id** refers to the type of accession of the sample, meaning the genetically-unique group that each sown seed belongs to. **phenotype_value** specifies the mean root length of each measured plant, and **phenotype_name** shows the treatment that each sample was put under, i.e. C (for Control), IAA, ABA, and CK.

The analysis first begins with descriptive statistics, calculating mean total root length (over all accessions within a treatment), standard error, and subsampling uncertainty, as well as skewness and excess kurtosis. The two last analyses were not included in the final results due to time limitations. The subsampling was not used as a main result, but rather as a check of how much the results vary when only a part of the dataset is used, as opposed to complete clustering of data.

The code also calculates the ratio of hormone measurements, resulting in relative total root length (hormone treatment in comparison to control):

$$R_{i,h} = \frac{TRL_{i,h}}{TRL_{i,\text{control}}}$$

with R denoting ratio, TRL with subscript i, h denoting hormone treated Total Root Length, and TLR with subscript *control* denoting control Total Root Length.

Lastly, Pearson's correlation was calculated to test how related the hormonal effects of each treatment were for each accession:

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}$$

Results and discussion

The first bar graphs simply show the distribution of total root lengths, respective to the number of accesssions per mm value.(Fig. 1) Control plants have the largest mean root length, lying at approximately 36.43 mm, with IAA and CK being very similar (35.08 and 35.46 mm respectively). IAA has the largest subsampling uncertaintly, which denotes that estimates across subsets of its data are less stable than initially perceived. ABA, on the other hand, has a range that is conspicuously shifted towards shorter lengths, with a

mean of 26.55 mm. This suggests that ABA may have a negative effect on the root length, and the strongest effect when compared to other hormonal treatments.

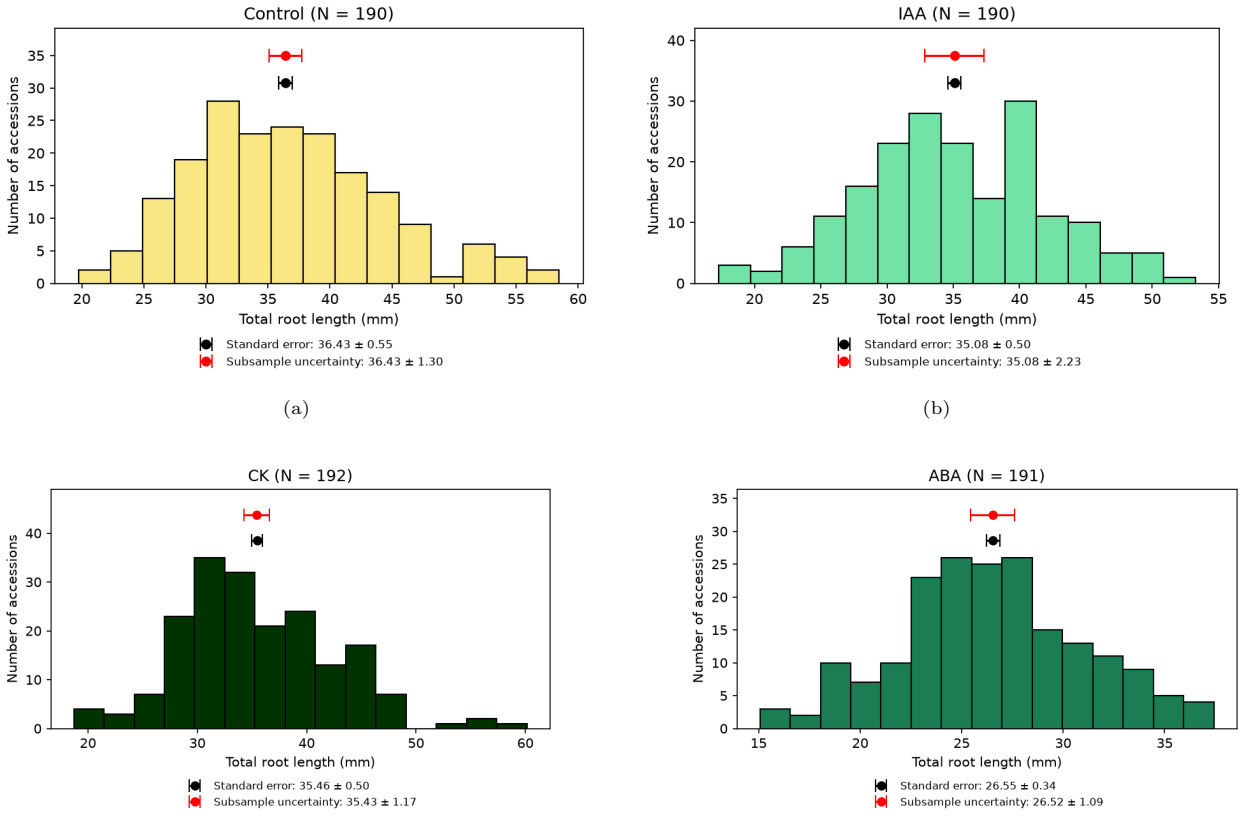


Figure 1: Bar graphs showing number of accessions per each root length value: Control (A), IAA(B), CK (C), and ABA (D).

The second bar graphs depict the comparisons of the overall mean root lengths (Fig. 2a) and overall relative root lengths (Fig. 2b). In both images, it is clear that ABA's root lengths are the shortest when compared to other treatments. This is also supported by the -29.0σ deviation for ABA in relative root length, which denotes that ABA values are 29 standard errors away from that of control (Fig. 2b). Herein, $\pm 3\sigma$ is considered significant change, so the value of -29 indicates statistical deviation and a large one at that. Compared to other σ values, ABA far exceeds any of them, so it can be determined that ABA has the strongest effect on root length, with IAA and CK depicting no significant influence (both $>3\sigma$).

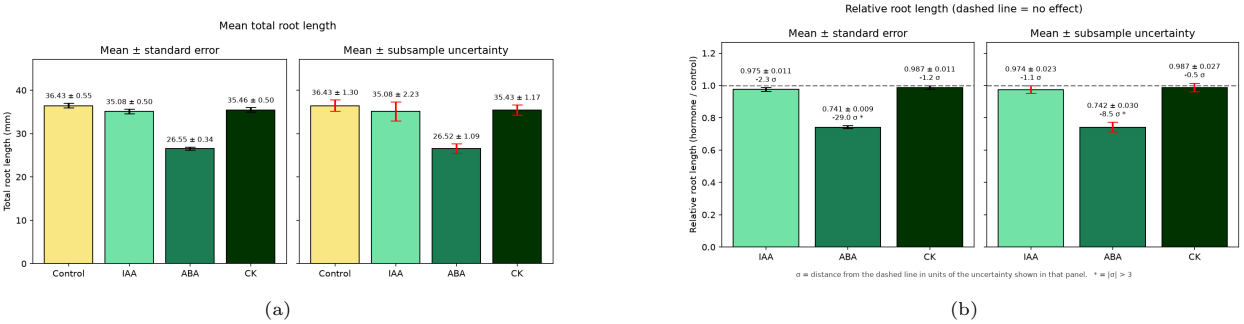


Figure 2: Bar graphs depicting comparison of mean root length (A) using overall mean and subsampling mean, with standard error and uncertainties respectively, and (B) using overall and subsample relative root length, with standard error and uncertainties respectively.

The last figures (Fig. 3) compare relative root lengths of the hormones to each other, in order to determine whether there is any correlation between the treatments. IAA vs CK have the largest r value at 0.61 ± 0.6; such a value shows that the two are quite correlated, since the closer the r value reaches positive 1, the more positively linear the relationship between 2 values is. Thus, it can be determined that IAA and CK are statistically correlated. Their standard deviation also stands at 10.6σ, which means that their correlation is 10.6 standard errors away from zero relation.

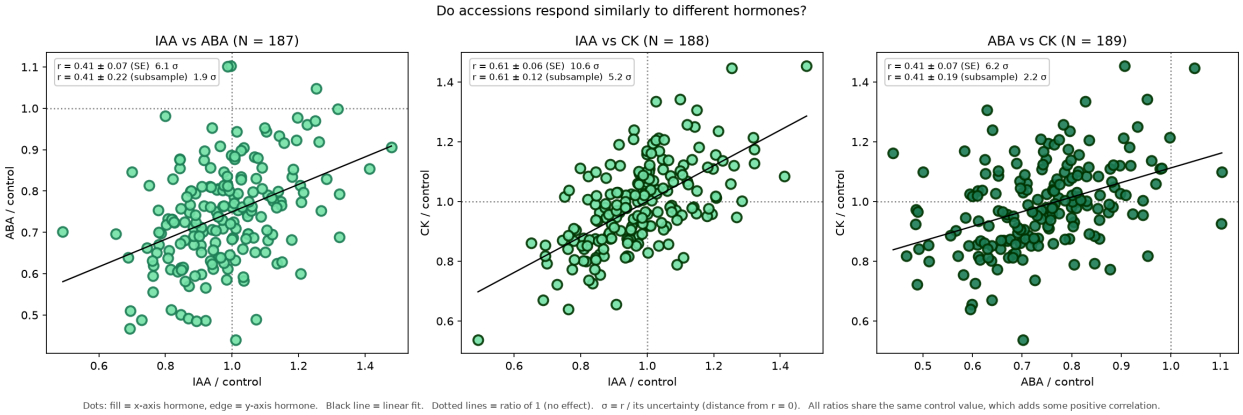


Figure 3: Scatter plot representing the relative root length of one hormone over that of another hormone per accession. Each dot represents a single accession, and the dotted lines (1.0 on both y and x-axis) represent the value of "no effect" in respect to control values. It is noted that all calculations are normalized by the same control values, so the observed correlations are skewed in part due to mathematical artifact. As such, true biological correlations may be lower than those depicted in the figures.

Altogether, it is confidently deduced that the effects of IAA and CK differ the least per accession root, with ABA having the strongest lone effect on root length values.

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